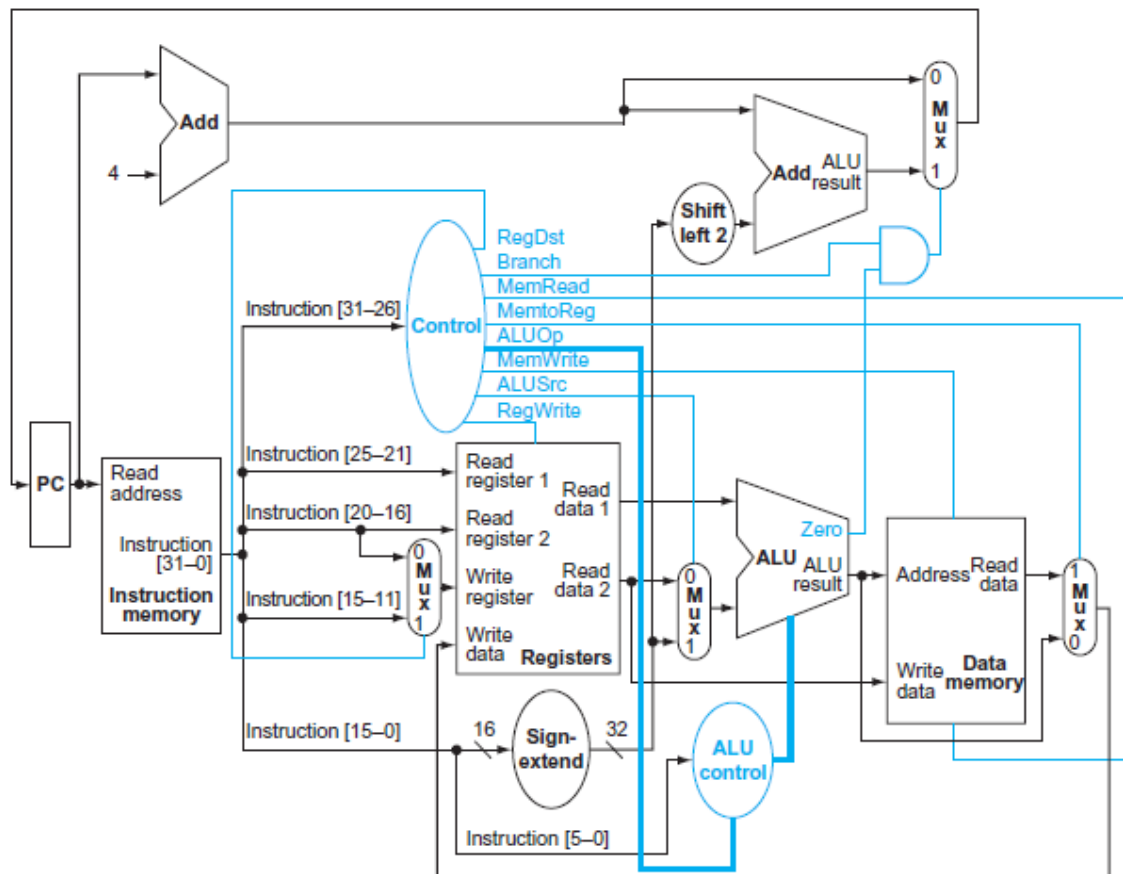


Department of Electrical and Computer Engineering
University of Wisconsin – Madison
ECE 552 Introduction to Computer Architecture
In-Class Exercise (09/26)

1. Consider the simple single-cycle MIPS processor shown below (from COD Figure 4.17).



Fill in the following table, specifying for each instruction whether the given **control signal** should be set to **0**, **1** or **X (don't care)**.

Answer:	ALUSrc	RegWrite	RegDst
sll	0	1	1
lw	1	1	0
sw	1	0	X
bne	0	0	X

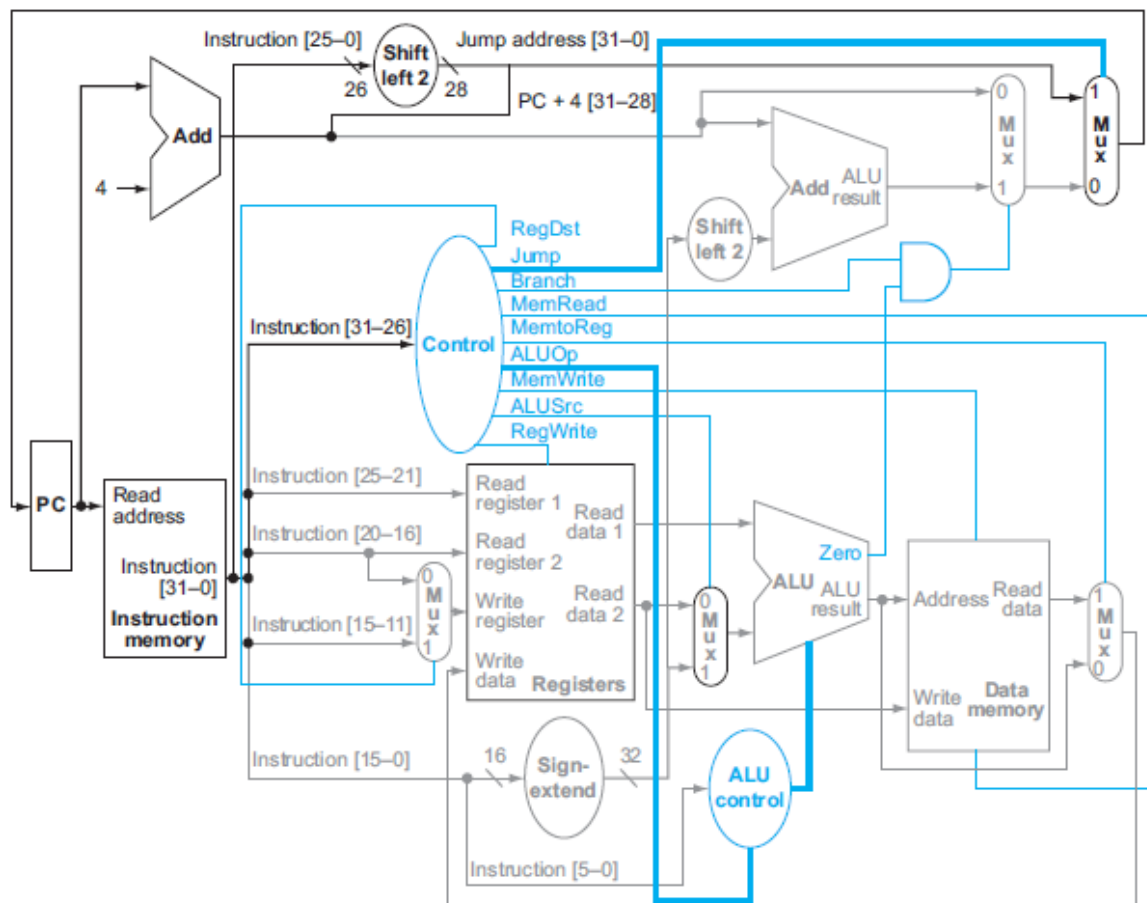
2. You are building a single-cycle processor (COD Figure 4.17) to support the MIPS **core instruction set**. Complete the assign statement below (in Verilog syntax) to properly set the **MemtoReg** control signal using only the **opcode (Instruction[31:26])** of the instruction word as input. Assume that you do **not** have to support any jump instructions (i.e., j, jal, jr) nor the Load Linked (ll)/Store Conditional (sc) instructions. Try to minimize to the fewest number of gates.

assign MemtoReg = _____;

Answer:

assign MemtoReg = Instruction[31];

3. We extend our single-cycle MIPS processor to support jump instructions as shown below (COD Figure 4.24).



Assume the following MIPS instruction is currently being executed on this processor:

add \$t8, \$t6, \$s5

- (a) What is the output of the **sign-extend** unit for this instruction? Provide your answer in 32-bit hexadecimal.

Answer:

**1111 1111 1111 1111 1100 0000 0010 0000
0xffffc020**

- (b) What is the output of the **jump shift left 2** unit (top left of COD Figure 4.24; not to be confused with the branch shift left 2 unit) for this instruction? Provide your answer in 28-bit hexadecimal.

Answer:

**0111 0101 0111 0000 0000 1000 0000
0x7570080**